

Hormonal therapy for menopause and breast-cancer risk by histological type: a cohort study and meta-analysis

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Summary

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Background Little information is available on how the risk of breast cancer associated with the use of hormone therapy for menopause varies by histological type. We aimed to describe such associations for eight histological types of breast cancer.

Methods Analyses are based on 1 031 224 postmenopausal women recruited in 1996–2001 into a nationwide UK cohort study, and followed for incident cancer and death. Relative risks associated with use of hormone therapy were estimated for eight histological types of breast cancer.

Findings During 3·6 million person-years of follow-up, 14 102 breast cancers were diagnosed, of which 13 782 (98%) had histological type recorded: 11 869 (86%) were invasive, including 8007 ductal, 1526 lobular, 365 mixed ductal–lobular, 492 tubular, 71 medullary, and 148 mucinous cancers; and 1913 (14%) were in situ, including 1443 ductal and 86 lobular cancers. The relative risks of invasive breast cancer in current users compared with never users of hormone therapy varied significantly according to tumour histology overall ($p < 0·0001$), for users of oestrogen-only therapy ($p = 0·0001$), and for users of oestrogen–progestagen therapy ($p < 0·0001$). The largest relative risks in current compared with never users of hormone therapy were seen for lobular (relative risk 2·25, 95% CI 2·00–2·52), mixed ductal–lobular (2·13, 1·68–2·70), and tubular cancers (2·66, 2·16–3·28). The relative risks for ductal and mucinous cancers were 1·63 (95% CI 1·55–1·72) and 1·58 (1·08–2·31), respectively. The risk of medullary cancer was not increased (0·74, 0·43–1·28). The relative risk of in-situ disease in current users compared with never users of hormone therapy also varied significantly according to histological type ($p = 0·03$), with a relative risk for lobular carcinoma in situ of 2·82 (1·72–4·63) and 1·56 (1·38–1·75) for ductal carcinoma in situ. The effects of hormone therapy on invasive ductal, lobular, and tubular cancer were generally greater for oestrogen–progestagen therapy than for oestrogen-only therapy, and were attenuated with increasing body-mass index (BMI).

Interpretation The risks associated with use of hormone therapy for menopause differ by histological type of breast cancer, and are substantially attenuated with increasing BMI.

Introduction

Evidence from randomised controlled trials and observational studies^{1–20} has shown that women taking hormone therapy for menopause are at an increased risk of breast cancer. Moreover, the risk of breast cancer associated with the use of hormone therapy is greater for users of combined hormone therapy than for users of oestrogen-only preparations.^{1,4,5,9,11,13,16} Less information is available about whether the effect of hormone therapy on breast cancer differs according to histological type. Several studies^{3,6,8–10,12–14,17–20} have assessed the effects of hormone therapy on specific histological types of breast cancer. Although most studies have suggested that use of hormone therapy is more strongly associated with lobular than with ductal cancer, their findings have not been entirely consistent. Few studies have had sufficient power to estimate reliably the risk of particular histological types of breast cancer according to the type of hormone therapy preparation used, or according to factors that modify the effect of hormone therapy, such as body-mass index (BMI). The aim of this report is to assess the relative risk of eight histological types of breast cancer associated with different types and patterns of postmenopausal hormone-therapy use in a large cohort of women in the UK.

Methods

Patients and procedures

1·3 million women aged 50–64 years were recruited into the Million Women Study in 1996–2001. All women completed a recruitment questionnaire on reproductive factors, sociodemographic factors, and other personal characteristics, including use of hormone therapy. A follow-up questionnaire was sent to all study participants around 3 years after recruitment to obtain updated information on menopausal status, use of hormone therapy, and other factors, with a response rate of 65%. Full details of the study design and methods have been described elsewhere,²¹ and both questionnaires can be viewed at <http://www.millionwomenstudy.org>. Study participants have been flagged on the UK National Health Service (NHS) Central Registers so that cancer registrations and deaths are notified routinely to the study investigators, including the date of the event and the code for the cancer site and morphology according to the 10th revision of the International Classification of Diseases (ICD10).²² All participants gave written informed consent to take part in the study, and ethics approval was provided by the Oxford and Anglia Multi-Centre Research and Ethics Committee.

	Women with breast cancer (n=13 782)
Invasive disease	11 869 (86%)
Ductal	8007 (58%)
Lobular	1526 (11%)
Mixed ductal-lobular	365 (3%)
Tubular	492 (3%)
Medullary	71 (1%)
Mucinous	148 (1%)
Other or unknown	1260 (9%)
In-situ disease	1913 (14%)
Ductal	1443 (10%)
Lobular	86 (1%)
Other or unknown	384 (3%)

Table 1: Distribution of histological types of incident breast cancer

The endpoint was defined as incident invasive breast cancer (ICD10 C50) or in-situ breast cancer (ICD10 D05). Women with incident invasive or in-situ breast cancer were classified further according to the histological type of their cancer, as defined by the ICD10 morphology code provided routinely by the NHS Central Registers. Women with invasive breast cancer were classified according to whether the cancer was ductal (ICD10-0 code 8500/3), lobular (ICD10-0 code 8520/3), mixed ductal-lobular (ICD10-0 code 8522/3), tubular (ICD10-0 code 8211/3), medullary (ICD10-0 code 8510/3), mucinous (ICD10-0 code 8480/3), or other (including invasive tumours not otherwise specified: all other ICD10-0 codes under ICD10 C50). Women diagnosed with in-situ disease were classified into three groups, corresponding to the codes used for invasive disease: ductal carcinoma in situ (ICD10-0 code 8500/2), lobular carcinoma in situ (ICD10-0 code 8520/2), or other in-situ disease (including in-situ disease not otherwise specified: all other ICD10-0 codes under ICD10 D05).

The variables for these analyses, including menopausal status and use of hormone therapy, were derived from information provided at baseline, and were updated with

follow-up information wherever possible. A woman's menopausal status was defined according to criteria outlined previously,¹¹ and women were eligible for inclusion in this analysis if they were defined as postmenopausal either at recruitment or during follow-up. From the first point at which a woman was defined as postmenopausal, time since menopause was assumed to increase with time, until this variable could be updated, or until the end of follow-up. Information recorded about use of hormone therapy at both recruitment and follow-up included: ever, current and past use; age at first and last use; total duration of use; and name of proprietary brand last used. Current and past use of hormone therapy, type of preparation last used, and total duration of use were updated, where possible with information from the follow-up questionnaire. For these analyses, current users of hormone therapy were classified into those who had last used either a combined preparation (ie, containing both oestrogen and progestagen), an oestrogen-only preparation, tibolone, or some other type of hormone therapy, on the basis of the hormonal constituents and formulation of each proprietary preparation of hormone therapy in the *British National Formulary*.²³

Statistical analysis

For every woman, person-years were contributed from the date of recruitment or from the time at which they were first classified as postmenopausal. Women diagnosed with invasive cancer other than non-melanoma skin cancer (ICD10 C44) before recruitment were excluded from all analyses, and women diagnosed with any cancer other than the cancer of interest (or non-melanoma skin cancer) during follow-up were censored at the date of diagnosis. Women continued to contribute person-years up to the earliest of: 48 months from last contact, registration for the breast cancer of interest, date of death, or the end of follow-up, which was defined as Dec 31, 2002. For Scotland, the end of follow-up was defined as Dec 31, 1999 because follow-up for this registry was incomplete beyond that date.

	Invasive ductal (n=8007)	Invasive lobular (n=1526)	Invasive tubular (n=492)	All women with breast cancer (n=13 782)
Age at entry, years	57.1 (4.1)	57.5 (4.1)	56.8 (4.2)	57.1 (4.2)
Upper-third of socioeconomic group, n (%)	2711 (34%)	546 (36%)	185 (38%)	4718 (34%)
Parity	2.0 (1.3)	2.0 (1.3)	2.0 (1.3)	2.0 (1.3)
Past use of oral contraceptives, n (%)	4470 (56%)	884 (58%)	265 (55%)	7742 (57%)
BMI, kg/m ²	25.9 (4.6)	25.4 (4.7)	24.7 (4.1)	25.5 (4.5)
Strenuous physical activity more than once per week, n (%)	2943 (38%)	589 (40%)	193 (40%)	5125 (39%)
Alcohol intake, g/week	36.4 (44.5)	38.4 (44.9)	42.0 (48.8)	36.6 (44.6)
Current smoker, n (%)	1605 (21%)	315 (22%)	135 (28%)	2761 (21%)
Age at birth of first child, years	24.2 (4.5)	24.7 (4.4)	24.3 (4.5)	24.3 (4.5)

Data are mean (SD) unless otherwise indicated, percentages exclude women with missing values.

Table 2: Characteristics of all women with breast cancer, and those with three main histological types of invasive breast cancer

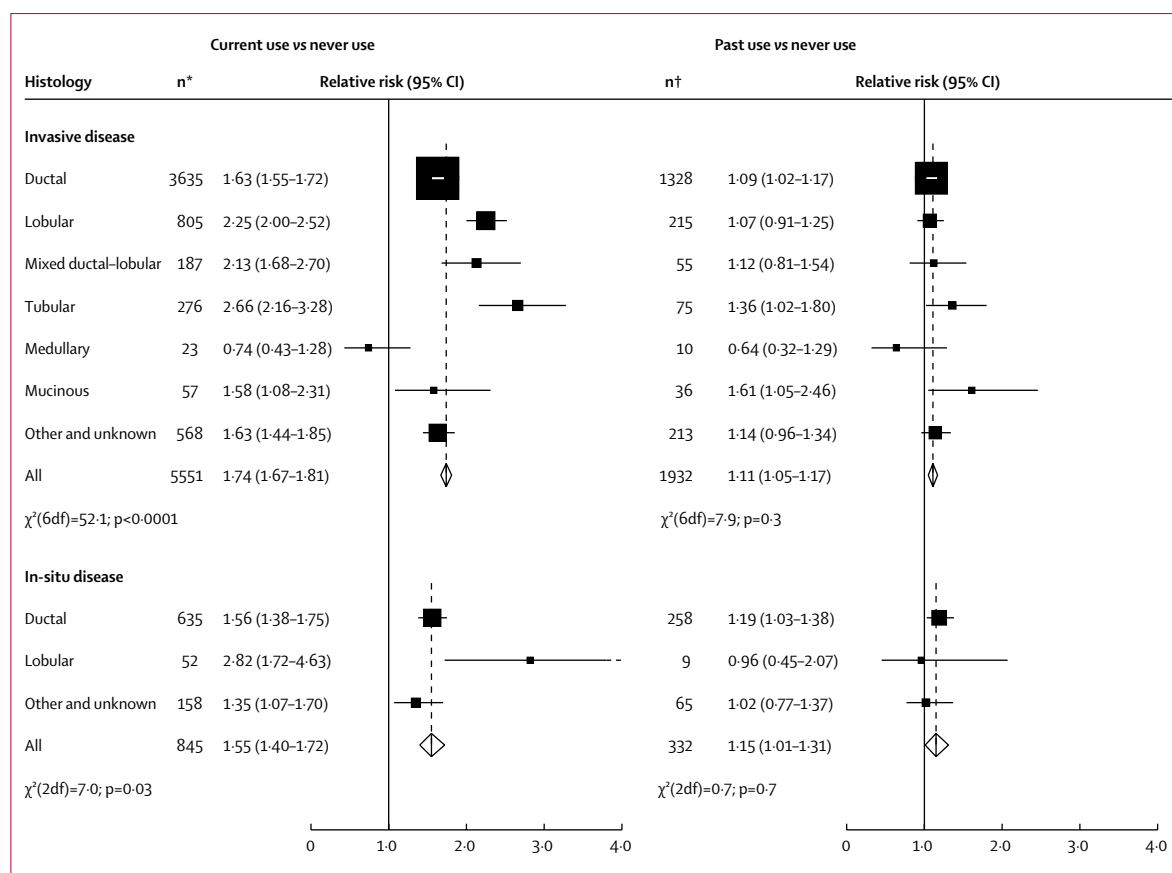


Figure 1: Relative risk of specific histological types of breast cancer in current users and past users of hormone therapy compared with never users
Relative risk stratified by age at entry, and adjusted for region, age at birth of first child, parity, time since menopause, deprivation index, BMI, and family history of breast cancer. df=degrees of freedom. *Cases in current users. †Cases in past users.

Proportional hazards models were used to estimate the relative risk of specific histological types of breast cancer in relation to use of hormone therapy, with attained age as the underlying time variable. Every histological category of breast cancer described above was treated as a single endpoint in a proportional hazards model, including pattern of hormone-therapy use as the main explanatory variable, stratified by age in years at entry (50–52, 53–55, 56–58, 59–61, and 62–64), and adjusted for broad geographical region (ie, ten regions covered by cancer registries that provided incidence data), age in years at birth of first child (<20, 20–29, ≥30, or not known), parity (defined as the number of full-term pregnancies; 0, 1–2, ≥3, or not known), years since menopause (<5, 5–9, or ≥10), quintiles of socioeconomic group on basis of deprivation index,²⁴ BMI (<25 kg/m², ≥25 kg/m², or not known), and history of breast cancer in mother or sisters, or both (no, yes, or not known). Time since last use of hormone therapy, type of hormone therapy preparation last used, total duration of hormone-therapy use, and time since menopause, were treated as categorical time-dependent variables. For current users of hormone therapy, the duration of current use at the time of

diagnosis of breast cancer was estimated from the duration of current use reported at the time of last contact and the time between last contact and the time of diagnosis, assuming that the women who developed breast cancer had the same pattern of use of hormone therapy before diagnosis as women who did not develop cancer (based on data for current users of hormone therapy who did not develop cancer and completed the follow-up questionnaire prior to Jan 1, 2003, an estimated 7% of oestrogen-only users, and 9% of oestrogen–progestagen users ceased use each year). Formal tests of heterogeneity in the effect of hormone therapy on breast-cancer risk in relation to histological type were done by use of a competing-risks approach. To assess the effect of the inclusion of women with missing information for some adjustment factors, analyses were repeated in women with known values for all adjustment variables. Sensitivity analyses based only on information reported at recruitment were also done to assess the effect of updating exposure information. All analyses were done with the statistical package Stata version 9.0.

Where results are presented in the form of plots, relative risks are represented by squares, the area of

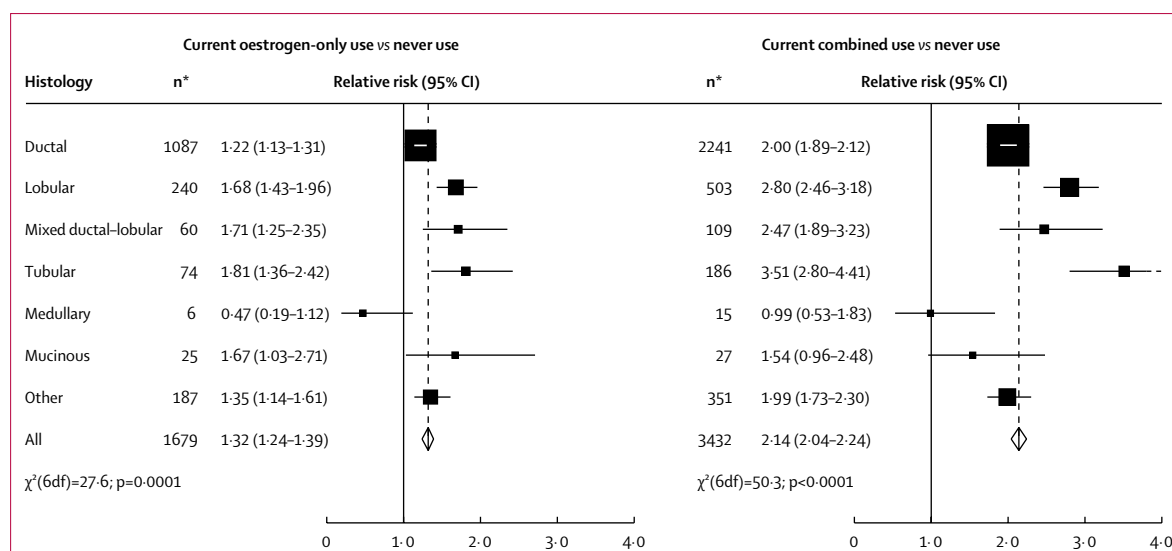


Figure 2: Relative risk of specific histological types of breast cancer in current users of oestrogen-only and oestrogen-progestagen hormone therapy compared with never users of hormone therapy

Relative risk stratified by age at entry, and adjusted for region, age at birth of first child, parity, time since menopause, deprivation index, BMI, and family history of breast cancer. df=degrees of freedom. *Cases in current users.

which is inversely proportional to the variance of the logarithm of the relative risk, and therefore, proportional to the amount of statistical information available for that estimate. 95% CIs are indicated by lines. In a meta-analysis of published data, summary relative risks are estimated by calculating the weighted average of the study-specific log relative risks, with weights proportional to the inverse of the variances of the study-specific log relative-risk estimates.

Role of the funding source

The sponsors of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

1031224 postmenopausal women were followed up for incident breast cancer over 3.6 million person-years. During this time, 14102 women were registered with incident breast cancer, and the median age at diagnosis was 59 years (IQR 55–62). The mean time for all women from date of last contact to end of follow-up was 2.7 years (SD 1.1). Breast cancers were diagnosed a mean of 1.2 years (SD 1.2) after the date of last contact. Information on histological type was provided by the NHS Central Registers for 13782 (98%) of these cancers, of which 11869 (86%) were invasive and 1913 (14%) were in situ (table 1). For those screen-detected cancers for which additional data on histological type were provided by the NHS Breast Screening Programme, the morphology codes showed excellent agreement with

those from the NHS Central Registers.²⁵ Table 2 shows characteristics of all women with breast cancer and those with the three main histological types of invasive breast cancer. No material differences were seen between women with invasive ductal, lobular, and tubular cancers in any of the characteristics assessed.

At the time of last recorded information on the use of hormone therapy, 455 338 (44%) women were never users, 373 232 (36%) were current users, and 202 654 (20%) were past users. In women who were current users of hormone therapy at last contact, 188 910 (51%) were using a combined oestrogen-progestagen preparation, 149 709 (40%) an oestrogen-only preparation, 22 006 (6%) tibolone, and 12 607 (3%) other, mixed, or unspecified types of preparation. At the time of this analysis, follow-up information was available for the first two-thirds of the study population with a median year of recruitment of 1998 (IQR 1997–1998). No appreciable differences were seen in use of hormone therapy, as recorded at recruitment, between women who did and did not return a follow-up questionnaire, although current smokers and women without any higher education were less likely than other women to respond. In the 392 341 (38%) women for whom follow-up information is included in this analysis, 328 602 (84%) reported the same status with respect to use of hormone therapy (ie, current, past, or never user) at both recruitment and follow-up, a mean of 2.8 years (SD 0.2) later. In addition, in women who specified type of hormone therapy last used at both recruitment and follow-up questionnaires, 122 039 (91%) of current users at recruitment reported last using the same type of hormone therapy at follow-up.

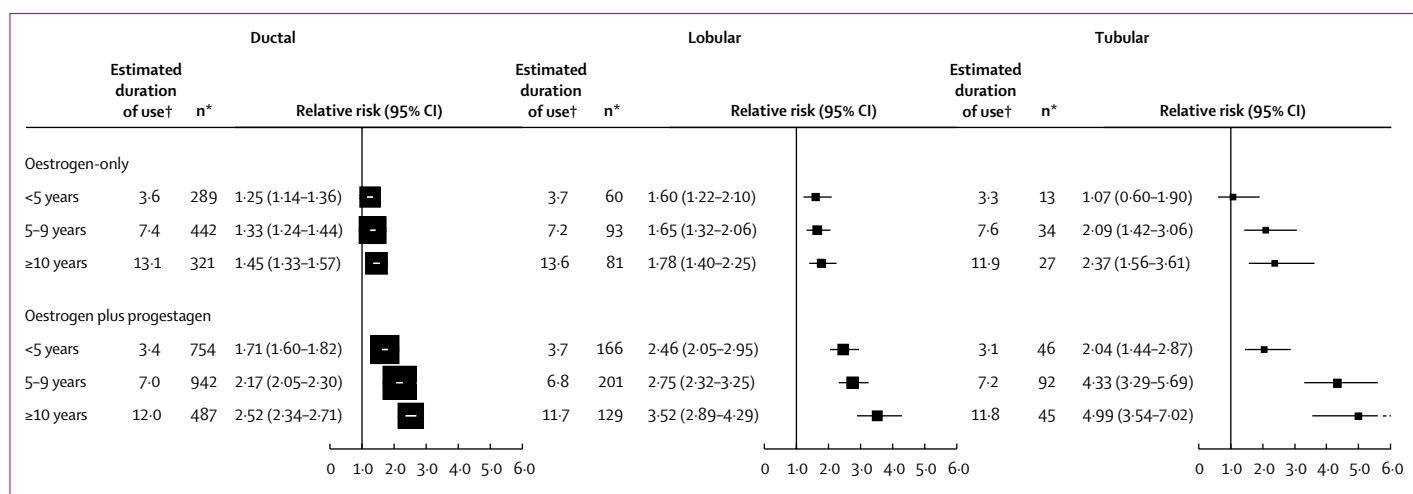


Figure 3: Relative risk of invasive ductal, lobular, and tubular breast cancer in current users of oestrogen-only and oestrogen-progestagen therapy compared with never users of hormone therapy by duration of use

Relative risk stratified by age at entry, and adjusted for region, age at birth of first child, parity, time since menopause, deprivation index, BMI, and family history of breast cancer. *Cases in current users. †Estimated years of use in cases at time of diagnosis (see methods).

Figure 1 shows the relative risk of each histological type of breast cancer shown in table 1 for current users and past users of hormone therapy compared with never users. The overall relative risk of invasive disease in current users versus never users of hormone therapy varied substantially according to histological type (figure 1). The relative risks for invasive ductal cancer, invasive mucinous cancer, and invasive cancers of other or unknown histology were all significantly increased in current users compared with never users and were all of the order of 1·6 (figure 1). By contrast, the relative risk for invasive lobular cancer was 2·25 and for invasive tubular cancer was 2·66. The risk of invasive medullary cancer was not increased (figure 1). The relative risk of mixed ductal-lobular cancer in current users compared with never users of hormone therapy was intermediate between that of the corresponding estimates for ductal and lobular cancer.

Relative risks in current users compared with never users of hormone therapy of specific histological types of in-situ disease differed significantly (figure 1). The pattern of relative risks in current users compared with never users of hormone therapy was similar to that for invasive disease in that the relative risk of lobular carcinoma in situ was 2·82 compared with 1·56 for ductal carcinoma and 1·35 for other in-situ disease.

For all types of invasive breast cancer combined, the relative risk was slightly higher in past users than in never users. When the overall relative risk in past users versus never users of hormone therapy was assessed against time since last use, the estimates in women who had ceased use less than 5 years ago, 5–9 years ago, and 10 or more years ago were 1·15 (1·07–1·24), 1·02 (0·92–1·13), and 0·95 (0·81–1·11), respectively. Therefore, the increase in risk in current users of hormone therapy declines after cessation of hormone-therapy use, and by 5 years after

ceasing use we found no significant excess risk. The relative risks in past users compared with never users of hormone therapy did not vary with the histological type of invasive breast cancer (figure 1). When analyses were based on only the information reported at recruitment, and when they were restricted to women with valid information for all adjustment variables, the results in figure 1 were essentially unchanged.

Figure 2 shows the relative risks in current users of oestrogen-only and oestrogen-progestagen therapy compared with never users of hormone therapy for every histological type of invasive cancer studied. There was significant heterogeneity in the relative risks in current users of both oestrogen-only and oestrogen-progestagen therapy, compared with never users of hormone therapy, by tumour histology. For both types of therapy, the relative risks were greatest for invasive lobular, mixed ductal-lobular, and lobular cancer. However, the relative risks were generally higher in current users of combined oestrogen-progestagen therapy compared with current users of oestrogen-only therapy. There was no increase in the risk of medullary cancer associated with current use of either type of therapy.

Figure 3 shows the relative risk in current users of oestrogen-only and oestrogen-progestagen therapy compared with never users of hormone therapy according to duration of use for invasive ductal, lobular, and tubular cancers. We found no significant difference in the trends in relative risk with duration of use of either type of hormone therapy for ductal, lobular, or tubular cancer (figure 3). In current users of tibolone, compared with never users of hormone therapy, the relative risks of invasive ductal (n=307), lobular cancer (n=62), and tubular cancer (n=16) were 1·47 (1·30–1·65), 1·81 (1·39–2·35), and 1·67 (0·99–2·80), respectively.

	<25 kg/m ²		25–29 kg/m ²		≥30 kg/m ²		Heterogeneity	p
	n*	Relative risk (95% CI)	n*	Relative risk (95% CI)	n*	Relative risk (95% CI)		
Ductal								
Oestrogen-only	490	1.46 (1.31–1.63)	369	1.12 (0.99–1.26)	177	1.01 (0.85–1.19)	χ ² (2df)=23.5	<0.0001
Oestrogen–progestagen	1165	2.28 (2.09–2.48)	730	1.99 (1.81–2.19)	266	1.63 (1.41–1.89)	χ ² (2df)=26.1	<0.0001
Lobular								
Oestrogen-only	113	1.94 (1.55–2.51)	87	1.27 (1.17–1.39)	33	1.10 (0.96–1.25)	χ ² (2df)=26.0	<0.0001
Oestrogen–progestagen	301	3.46 (2.88–4.16)	147	2.35 (1.94–2.85)	41	1.66 (1.48–1.86)	χ ² (2df)=94.8	<0.0001
Tubular								
Oestrogen-only	41	2.45 (1.63–3.70)	31†	1.43 (0.94–2.19)†			χ ² (1df)=3.16	0.08
Oestrogen–progestagen	114	4.39 (3.16–6.09)	61†	2.65 (1.87–3.76)†			χ ² (1df)=4.26	0.04

Relative risk stratified by age at entry, and adjusted for region, age at first birth, parity, time since menopause, socioeconomic status based on deprivation index, and family history of breast cancer. *Cases in current users. †Number of cases and relative risk are combined for women with BMI 25–29 kg/m² and ≥30 kg/m².

Table 3: Relative risk of invasive ductal, lobular, and tubular breast cancer in current versus never users of hormone therapy according to BMI and type of hormone therapy

Table 3 shows the relative risks of invasive ductal, lobular, and tubular cancer in current users compared with never users of hormone therapy according to BMI. For ductal and lobular cancer, the relative risks for both oestrogen-only and combined hormone therapy were significantly attenuated with increasing BMI. For tubular cancer, statistical power was low and the effect of current use was significantly attenuated with increasing BMI for combined therapy only. For ductal cancer, the extent of the difference in risk was such that although the relative risk in current users of combined therapy compared with never users of hormone therapy was still significantly greater than unity in overweight and obese women, no significant effect of oestrogen-only treatment on risk in overweight or obese women was found (table 3).

Discussion

In this large cohort study, we have shown that use of hormone therapy has varied effects on different histological types of breast cancer. Generally, we found that invasive lobular and tubular cancers are more strongly associated with current use of hormone therapy than is invasive ductal cancer.

A meta-analysis of the findings of ten previous studies^{3,8–10,12–13,17–20} that have assessed the effects of specific types of hormone-therapy use on the risk of different histological types of breast cancer, and the findings presented here are shown in figure 4. There were some minor differences between studies in the categorisations of tumour type and definition of hormone-therapy use. In particular, ductal carcinoma was defined in slightly different ways. We used a narrow definition of ductal cancer for both invasive and in-situ disease (ICD10-0 codes 8500/3 and 8500/2, respectively), which represents a more homogeneous group of cancers than would otherwise be the case and is in line with the codes used in other studies.^{3,13,19} Despite these slight differences in the definition of histological type, there is consistent evidence of a greater effect for both oestrogen-only and combined

hormone therapy on the risk of lobular and tubular cancers compared with ductal cancer. Few previous studies have had sufficient numbers of cases to show reliably such differences in histological type. Evidence from previous studies on the risk of tubular cancer is especially sparse. Although most studies found a greater effect of hormone therapy on lobular and tubular cancer than on ductal cancer, estimated relative risks of ductal cancer associated with both oestrogen-only hormone therapy and combined hormone therapy varied substantially (figure 4). Of the ten studies with data for the effect of oestrogen-only hormone-therapy use on the risk of ductal cancer, six found no significant effect on risk,^{3,8,10,12,13,17} two (including ours) a significant increase in risk,⁹ and two a significant decrease in risk.^{18,19} Furthermore, in the nine studies in which comparison of the risks of ductal cancer associated with oestrogen-only and combined hormone therapy was possible, the same three studies that reported the lowest relative risks for oestrogen-only therapy^{3,12,19} also reported the lowest relative risks for combined hormone therapy. The variability in the effect of either type of hormone therapy on ductal cancer is probably not attributable to differences in the specific types of preparations used across studies because no variation has been seen on the overall risk of breast cancer by type or dose of oestrogen or progestagen.¹¹

Several studies^{2,4,5,8,9,11,17} have found that the effect of hormone therapy on overall risk of breast cancer decreases with increasing BMI. One previous study²⁶ assessed the effect of hormone-therapy use by BMI in 623 women with ductal cancer and in 185 with lobular cancer. Although the researchers reported no significant interaction between hormone therapy and BMI for either type of breast cancer, the study had little power. Our study, which included 8007 women with ductal cancer, 1526 with lobular cancer, and 492 with tubular cancer, clearly shows that the effects of hormone therapy on each of these types of invasive cancer are sensitive to BMI, but the effect of oestrogen-only hormone therapy on ductal

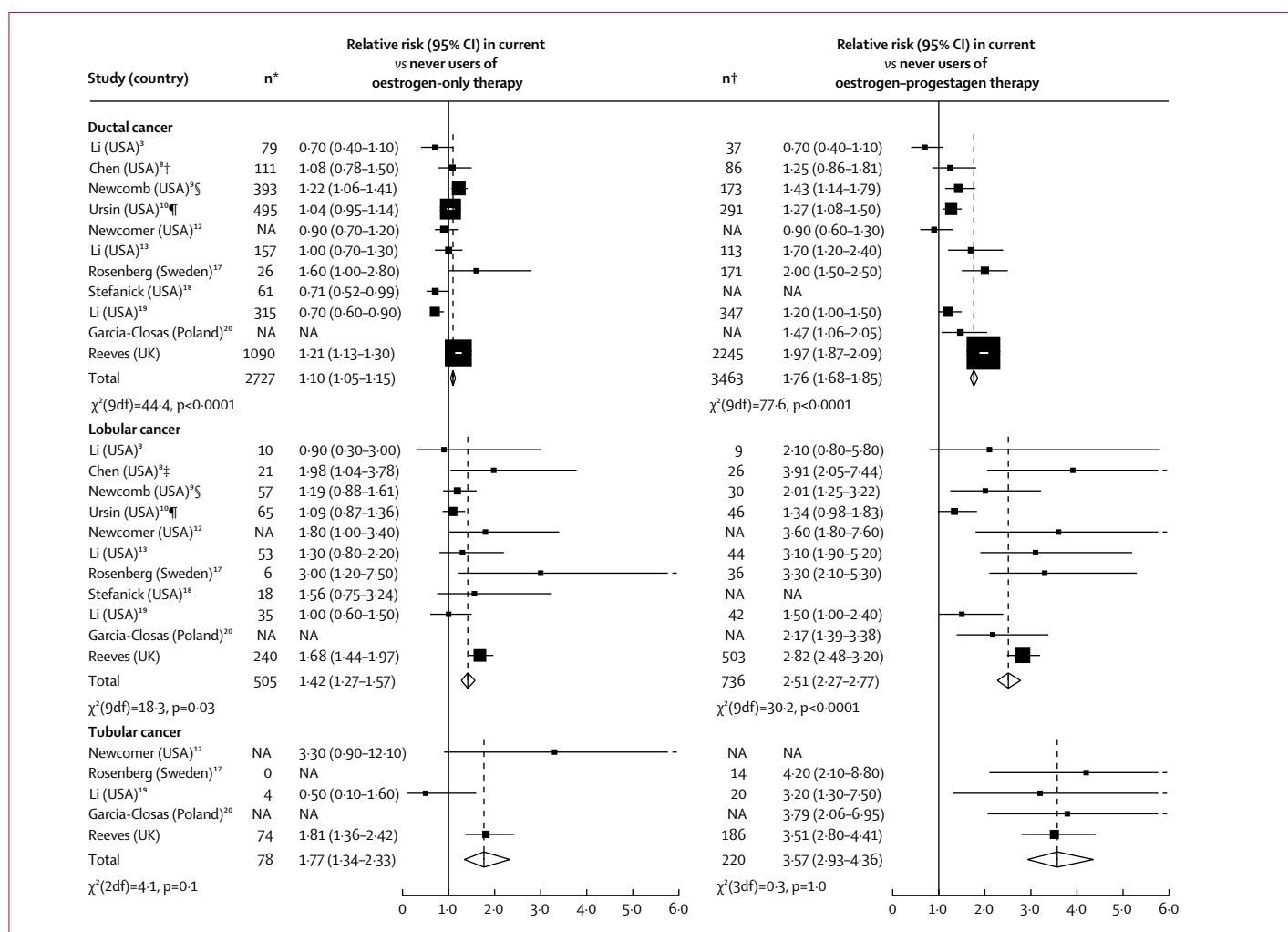


Figure 4: Meta-analysis of published results on the relation between use of oestrogen-only and oestrogen-progestagen hormone therapy and specific histological types of invasive breast cancer

NA=data not available. df=degrees of freedom. *Cases in users of oestrogen-only therapy. †Cases in users of oestrogen-progestagen therapy. ‡Relative risks of lobular cancer and all non-lobular cancer (mainly ductal carcinomas [89%]). §Relative risks in hormone therapy users, rather than current hormone-therapy users, compared with never users (60% of all oestrogen-only hormone-therapy users and 86% of all combined users were current users). ¶Relative risks in women with 5 years' duration of hormone-therapy use compared with never users.

cancer differs qualitatively according to BMI: use of oestrogen-only therapy increases the risk of ductal cancer in women who are not overweight, but shows no significant association with risk in overweight or obese women. Differences in the distribution of BMI between European and US women are well documented—eg, the proportion of hormone-therapy users who were obese was 18% in the Million Women Study, compared with 45% in the Women's Health Initiative trial.¹⁶ Therefore, a lower average BMI in women from Europe than in women from the USA could have contributed to the higher relative risks of ductal cancer associated with use of oestrogen-only therapy that were seen in our study and the only other European study with relevant data¹⁷ compared with those from the USA.

Two studies not included in figure 4 have shown the effects of any type of hormone therapy on the risk of

different types of breast cancer. The researchers of the Swedish cohort study⁶ reported relative risks of ductal, lobular, and tubular cancer in current users compared with never users of any type of hormone therapy of 1.25 (0.76–2.07), 4.38 (1.6–12.0), and 4.81 (1.37–16.8), respectively. In the other, a Danish cohort study,¹⁴ the corresponding relative risks for ductal and lobular cancer were 2.10 (1.64–2.70) and 3.53 (1.94–6.41), respectively. Although these estimates are difficult to compare with those relating to the use of specific types of hormone therapy presented in figure 4, they nonetheless support a greater effect of hormone-therapy use on the risk of invasive lobular and tubular cancers than on the risk of ductal cancer.

Compared with other histological types of breast cancer, lobular and tubular cancers are more likely to be positive for oestrogen and progesterone receptors, and

possibly for other hormone receptors.^{27,28} By contrast, proportionally fewer medullary cancers are positive for oestrogen or progesterone receptors than other types of breast cancer.²⁸ A review²⁹ of the published evidence did not suggest that hormone therapy is mainly associated with hormone-receptor-positive disease, but the patterns of risk seen here in relation to histological type are consistent with such a hierarchy of hormonal dependence. The apparent absence of an effect of overall hormone-therapy use on the risk of medullary cancer found here is especially important to this hypothesis. To our knowledge, only one other study¹⁹ has assessed the effect of hormone therapy on this type of breast cancer, and although the results are consistent with those presented here (relative risks associated with current use of oestrogen-only and combined hormone therapy of 0.5 [0.2–1.6] and 0.8 [0.2–3.0], respectively), they were based on a small number of cases. Although our study is the largest to investigate the effects of hormone therapy on specific histological types of breast cancer, the findings for some of the rarer types, especially medullary cancer, are still based on small numbers of cases, and, therefore, need to be confirmed in other studies.

We are not aware of any previous study that has assessed the effects of hormonal therapy on ductal and lobular carcinoma in situ. Most in-situ disease is detected at screening, so estimates of the effect of hormone-therapy use on the risk of in-situ cancer are prone to bias because mammography screening patterns might differ between hormone-therapy users and non-users. In our study, all women attended routine screening at the time of recruitment and were followed up for any incident cancer (irrespective of whether they were detected at screening or during intervals between screening), therefore, such potential biases were minimised. Under this design, the relative risks for current users versus never users of hormone therapy were significantly elevated for both ductal carcinoma in situ and lobular carcinoma in situ, but the extent of the excess risk was greater for lobular carcinoma in situ.

The Million Women Study is appropriate for the investigation of the effects of hormone therapy on specific types of breast cancer owing to its large sample size and the high quality of the information on tumour characteristics. Analyses presented here used histology information routinely provided by the NHS Central Registers. For women whose cancers are detected at routine screening, independent information on histological type is also available from the NHS Breast Screening Programme. Comparison of histological information provided by the NHS Central Registers and the NHS Breast Screening Programme in this subgroup of women has shown excellent agreement;²⁵ therefore, histological information provided by the NHS Central Registers is sufficiently reliable for use in epidemiological studies. Information on hormone-therapy use obtained at baseline was, where possible, updated from information

obtained through a follow-up questionnaire. For women included in the analysis, the mean time from last contact to end of follow-up was 2.7 years (SD 1.1), and the mean time from last contact to diagnosis was 1.2 years (SD 1.2) in women with breast cancer. Irrespective of whether exposure information had been updated, women remained in the analysis for a maximum of 48 months after the last date at which hormone-therapy use was recorded. Although the status of hormone-therapy use will have changed since the date of last contact for a small proportion of women, this should not unduly affect results that specifically relate to differences in breast-cancer risk according to histological type.

In summary, our findings show that the risks for invasive lobular and tubular cancers associated with current use of both oestrogen-only and combined oestrogen–progestagen hormone therapy are higher than those for invasive ductal cancer. Across histological types, the relative risks associated with combined hormone therapy are generally higher than those associated with oestrogen-only therapy. The effect of hormone therapy on ductal, lobular, and tubular cancer decreases as BMI increases, and differences in the distribution of BMI between populations might have contributed to the observed variability in the published data on the risk of invasive ductal cancer with hormone-therapy use. The clinical implications of these findings are, as yet, unclear. Future work should focus on assessing the extent to which the differences recorded here according to histological type are accounted for by differences in hormone-receptor status, or other tumour characteristics.

Conflicts of interest

We declare no conflicts of interest.

Contributors

All authors took part in the design and implementation of the study, and read and approved the final report.

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Collaborating UK NHS breast-screening centres

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